

Nelly Barret

ASSISTANT PROFESSOR AT INSA LYON AND LIRIS

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Research interests

I work in the broad field of data management, and more precisely my research interests focus on heterogeneous and/or multimodal data, data integration systems, and data exploration. Recently, I also added Federated Learning to my research interests. The projects that I worked on have always been part of multidisciplinary contexts, including urban planning, journalism, and health, and have been exemplified on large, complex, and often real-world data.



Academic positions

2025 - now	Assistant Professor INSA Lyon & LIRIS (FR)
2024 - 2025	Postdoctoral researcher Politecnico di Milano (IT)
2021 - 2024	PhD student Institut Polytechnique de Paris & Inria Saclay (FR)
2020 - 2020	Master intern LIRIS (FR)
2018 - 2018	Bachelor intern LIRIS (FR)



Education

2021 - 2024	PhD degree Institut Polytechnique de Paris
2018 - 2020	Master's degree Université Claude Bernard Lyon 1
2015 - 2018	Bachelor's degree Université Claude Bernard Lyon 1



Research visits

Institut de Recherche en Informatique de Toulouse (Oct. 2025 (2 days))



Awards

PhD thesis award (Oct. 2024)



Research projects

BETTER: real-world health-data distributed analytics research platform (Apr. 2024 – July 2025), CONNECTIONSTUDIO: user data exploration of heterogeneous data (March 2023 – March 2024), PATHWAYS: find interesting entity-to-entity paths in heterogeneous data (May 2022 – March 2024), ABSTRA: Entity-Relationship summaries from semi-structured data (Jan. 2021 – March 2024), PREDIHOOD: supervised prediction of the environment of neighborhoods (Feb. 2020 – July 2020), GEOALIGN: matching and merging geographic entities (Jan. 2019 – June 2019), VIZLIRIS: recommending and clustering French neighborhoods (May 2018 – July 2018),



Research software and tools

I-ETL (Apr. 2024 – July 2025), ConnectionStudio (March 2023 – March 2024), PathWays (May 2022 – March 2024), Abstra (Jan. 2021 – March 2024), Predihood (Feb. 2020 – July 2020), GeoAlign (Jan. 2019 – June 2019), VizLIRIS (May 2018 – July 2018)



Working groups

Commission égalité & parité femme-homme (Nov. 2025 – now), Diversity, Equity and Inclusion (Apr. 2023 – now), FAIRification of genomic annotations (Dec. 2024 – now)



Publications

My complete list of publications is available on my [ORCID record](#).



Professional service

Conference and workshop organizer: CORIA-TALN (Conférence en Recherche d'Information et Applications - Traitement Automatique des Langues Naturelles): 2027

Conference and workshop chair: CMLS (Conceptual Modeling for Life Sciences workshop): 2025 (workshop chair), 2026 (workshop chair), EDBT/ICDT Joint Conference (Extending DataBase Technology): 2027 (Diversity & Inclusion Chair),

Conference program committee member: ICDE (International Conference on Data Engineering): 2025 (reviewer), CAiSE (Conference on Advanced Information Systems Engineering): 2025 (reviewer), CIDR (Conference on Innovative Data systems Research): 2023 (reviewer), CIKM (International Conference on Information and Knowledge Management): 2026 (pc member), EDBT (Extending DataBase Technology): 2023 (reviewer), iSAILS (Information Systems and AI for Life Sciences workshop): 2025 (pc member), TGD (Transforming Graph Data workshop): 2026 (pc member), ICWE (International Conference on Web Engineering): 2021 (reviewer), RCIS (Research Challenges in Information Science): 2026 (forum pc member), ADBIS (Advances in Databases and Information Systems): 2026 (pc member), 2025 (pc member), DBKDA (Databases, Knowledge, and Data Applications): 2025 (pc member), BDA (Gestion de Données – Principes, Technologies et Applications): 2025 (demonstrations reviewer), 2026 (pc member),

Journal review board member: VLDB Proceedings (Proceedings of the Very Large DataBases volume): 2027

Journal reviewer: BMC Bioinformatics: 2025, BMC Medical informatics and Decision Making: 2024, GigaScience: 2025, PLoS One: 2024, Scientific Reports: 2024, JDIQ (Journal of Data and Information Quality): 2025, CMC (Computers, Materials and Continua): 2025, JOSS (Journal of Open Source Software): 2025,



Institutional responsibilities

2026 - now	GRID500 platform LIRIS referent main maintainer
2025 - now	DRIM webpage maintainer main maintainer
2024 - 2025	LinkedIn page maintainer main maintainer
2023 - 2024	PhD representative main representative
2021 - 2024	Team seminar organizer main organizer
2021 - 2021	Internship application website contributor active contributor



Talks

My complete list of talks on my [webpage](#).



Teaching responsibilities

Recurring courses at INSA Lyon: ISN1: Informatique et Société Numérique (fall, engineer 1st year): algorithmics, Python development | SN2: Informatique et Société Numérique (spring, engineer 1st year): algorithmics, functional decomposition, introduction to networks, guided hands-on project | SN4: Informatique et Société Numérique (spring, engineer 2nd year): object-oriented programming, graphical interfaces, code versioning, hand-on project | SOL: Systèmes et Outils Logiciels (fall, engineer 1st year): Microsoft Office and LibreOffice suite, basic Linux commands | P2I: Projets Pluridisciplinaires d'Initiation à l'Ingénierie (spring, engineer 2nd year): mecatronic and robotics hand-on project with a part dedicated to programming and communication between objects

Involvement in INSA: ISN2: new hands-on project topic | SN2: network course redesign | CGA: member of the Coordination and Management Committee (Comité de Gestion et d'Animation) | First year admission interviews for INSA Lyon

Training: Training on doctoral supervision (18 hours) | Training on student evaluation (4 hours) | Training on APC -- Competency-based approach (2 days)

Previous courses: CSE203: CS project (spring 2024, Ecole Polytechnique, bachelor 2nd year, in English) | CSE204: Machine Learning (spring 2023, Ecole Polytechnique, bachelor 2nd year, in English) | INF411: bases de programmation et d'algorithmique (fall 2022, Ecole Polytechnique, engineer 2nd year, in French) | INF371: mécanismes de la programmation orientée objet (spring 2022, Ecole Polytechnique, engineer 1st year, in French)



Advising

Engineer: Côme Périn on "Apprentissage multimodal au service de la santé des conducteurs" (April-Dec. 2026, ENTPE Vaulx-en-Velin)

Master students: Inès Boudjeli on "Nettoyage et sélection automatique de données pour l'apprentissage machine" (spring 2026, INSA Lyon, master 2nd year) | Ahmed Bel Hadj Youssef on "Construction automatique de domaines dans les data meshes" (fall 2025, INSA Lyon, engineer 5th year) | Antoine Gauquier on "Path exploration of ConnectionLens graphs" (summer 2022, IMT Nord-Europe, master 1st year)

Bachelor students: Shay Pripstein on "Target data acquisition in open repositories" (winter 2023, École Polytechnique, bachelor 3rd year) | Tudor Enache on "From simple to property graphs" (winter 2023, École Polytechnique, bachelor 3rd year) | Nikola Dobricic on "Graph queries for abstractions" (summer 2023, Ecole Polytechnique, bachelor 2nd year) | Jia-Jean Law on "Optimization of entity-to-entity data paths" (winter 2022, Ecole Polytechnique, bachelor 3rd year)